Level 3



TWO BEAMS BUILD. THREE BEAMS BURN.

```
FAB-42 > firmware 2.0 .. INSTALLED
FAB-42 > beam power .... 50%
```

A single laser beam turned out to be a blunt instrument: everything it touched became matter, wanted or not.

So, Dr. Ångström halved the power. One beam alone now passes through the board without leaving a trace. Only where two beams cross is there enough energy to create matter.

Where three or more beams meet, there is far too much: any matter in that cell is destroyed in a tiny, very expensive puff of smoke.

Simulate the newly calibrated emitters.

Multiple emitters are now used at the same time. They are different from the previous level.

- Now only when 2 laser beams hit the same cell, matter is created
- When 3 or more laser beams hit the same cell, matter is destroyed
- You are given a list of boards with some matter already on it and up to 6 emitters
- **Determine the state of the board after matter is created and destroyed**
- The directions of the emitters work identical to the previous level
- Laser beams interact only with `‘0’` and `‘M’`, not with `‘-’`, `‘*’` or `‘1’-’6’`

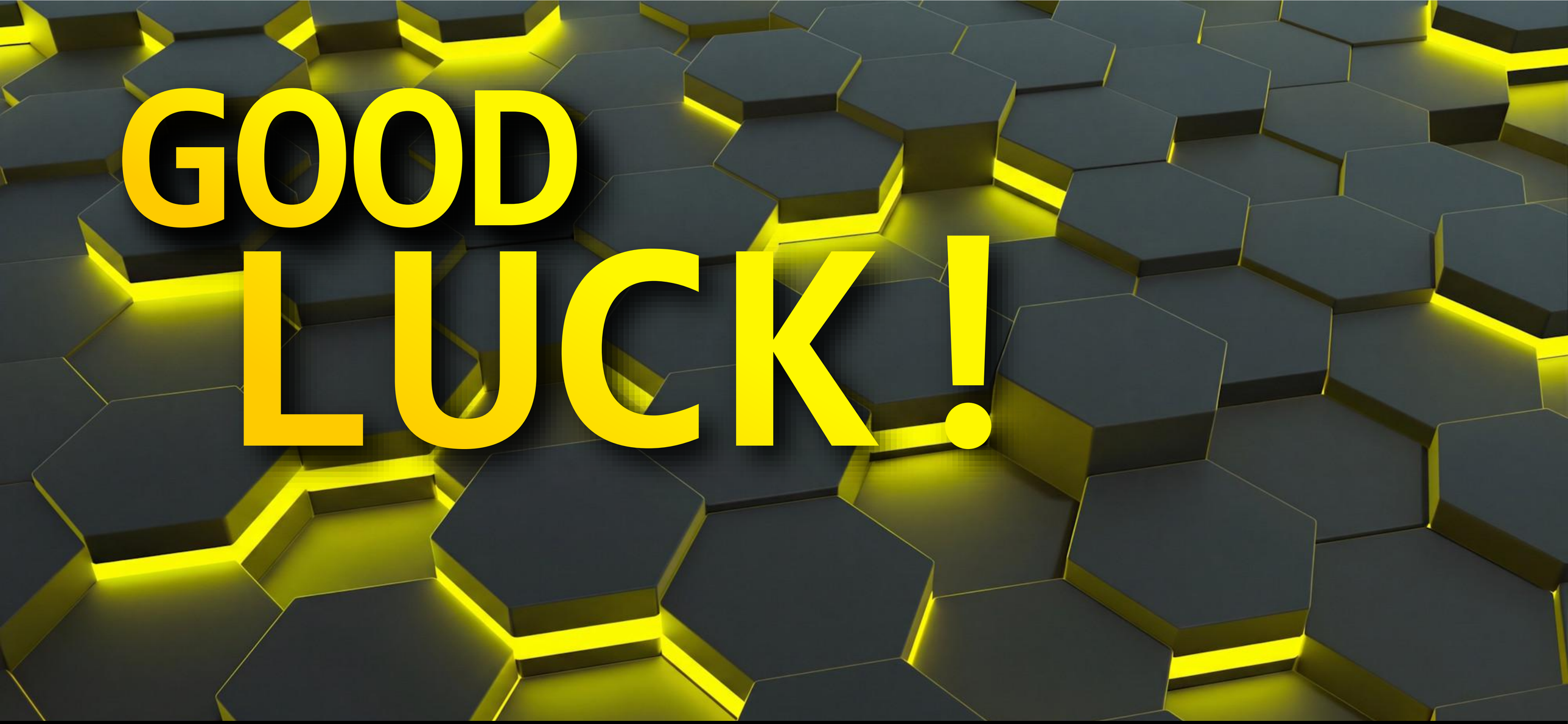


Input

Name		Description	Example
N		Number of boards	3 3 7 13 ---2-*-*-3--- --*-M-M-O-*-- -*-O-O-O-O-*- 1-O-O-M-O-O-* -*-O-M-O-O-4- --*-O-O-O-*-- ---*-*-*-*--- 4 9 17 ----2-*-*-3-*---- ---*-O-M-M-M-*--- ...
repeated N times	Board dimensions	The radius of the board, the number of rows, and the number of columns, separated by spaces	
	Board	The character representation of the board	
	Empty line between boards		

Output

Name		Description	Example
repeated N times	Board	The character representation of the board *You can print the border with emitters or without. Both will be judged as correct.*	---2-*-*-3--- --*-M-M-O-*-- -*-O-O-O-O-*- 1-O-O-O-O-O-* -*-O-M-M-O-4- --*-O-O-O-*-- ---*-*-*-*--- ----2-*-*-3-*---- ---*-O-M-M-M-*--- ...
	Mandatory empty line between boards		



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